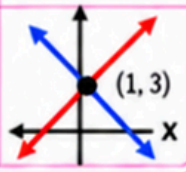
Algebra Systems of Equations Cheat Sheet

Unit Focus: *solving systems by graphing, substitution, and elimination*

1 KEY VOCABULARY

system of equations	Two or more equations with the same variables.	$\begin{cases} y = 2x + 1 \\ y = -x + 4 \end{cases}$
solution	An ordered pair that makes all equations in the system true.	$(1, 3)$
ordered pair	A pair of numbers written as (x, y) that represents a point on a coordinate plane.	$(-2, 5)$
intersect	The point where two lines cross. The intersection is the solution of the system.	

2 WHAT A SOLUTION MEANS ON A GRAPH

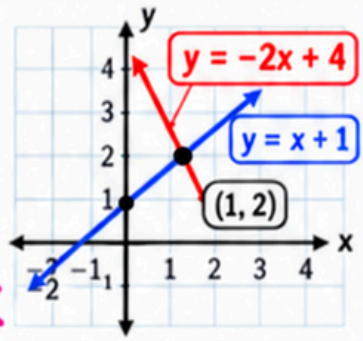
The solution of a system of equations is the point where the graphs of the equations intersect. The ordered pair at the intersection makes both equations true.

Example: $\begin{cases} y = x + 1 \\ y = -2x + 4 \end{cases}$

Step 1: Graph both equations.

Step 2: Find the point of intersection.

Step 3: The ordered pair is the solution.



Answer: $(1, 2)$

3 SOLVING BY GRAPHING

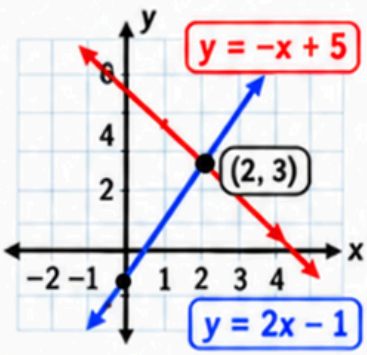
Graph both equations and find the point of intersection.

Example: $\begin{cases} y = 2x - 1 \\ y = -x + 5 \end{cases}$

Step 1: Graph $y = 2x - 1$ and $y = -x + 5$.

Step 2: Find the intersection point.

Step 3: Write the solution as an ordered pair.



Answer: $(2, 3)$

4 SOLVING BY SUBSTITUTION

Solve one equation for one variable, substitute into the other equation, and solve.

Example: $\begin{cases} y = 3x + 1 \\ 2x + y = 11 \end{cases}$

Step 1: Solve the first equation for y: $y = 3x + 1$

Step 2: Substitute into the second equation: $2x + (3x + 1) = 11$

Step 3: Solve for x:

$$5x + 1 = 11$$
$$5x = 10$$
$$x = 2$$

Step 4: Substitute back to find y: $y = 3(2) + 1 = 7$

Answer: $(2, 7)$

5 SOLVING BY ELIMINATION

Add or subtract the equations to eliminate one variable.

Example: $\begin{cases} 3x + 2y = 14 \\ 5x - 2y = 6 \end{cases}$

Step 1: Add the equations to eliminate y.

$$\begin{array}{r} 3x + 2y = 14 \\ +\ 5x - 2y = 6 \\ \hline 8x = 20 \\ x = 2 \end{array}$$

Step 2: Substitute $x = 2$ into an equation to find y.

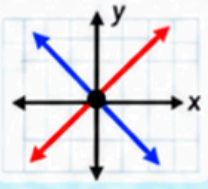
$$\begin{aligned} 3(2) + 2y &= 14 \\ 6 + 2y &= 14 \\ 2y &= 8 \\ y &= 4 \end{aligned}$$

Answer: $(2, 4)$

6 TYPES OF SOLUTIONS

A system of equations can have **one solution**, no solution, or infinitely many solutions.

One Solution (intersecting lines)



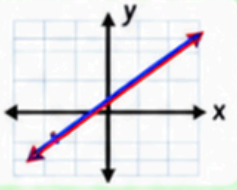
Exactly one point of intersection. Example: (1, 2)

No Solution (parallel lines)



Lines never intersect. No ordered pair satisfies both equations.

Infinitely Many Solutions (same line)



Both equations represent the same line. Every point on the line is a solution.

7 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- A solution must make both equations true.
- Check your answer by substituting into both equations.
- When graphing, the solution is the point of intersection.
- Use substitution or elimination to solve exactly.

COMMON MISTAKES:

- Graphing incorrectly (plotting points wrong).
- Forgetting to substitute into the other equation.
- Making sign errors when adding/subtracting.
- Thinking parallel lines have a solution.
- Thinking the same line has only one solution.

Algebra Systems of Equations Cheat Sheet

More Practice: *substitution, elimination, and applications*

1 SOLVING BY SUBSTITUTION

Solve one equation for a variable, substitute into the other equation, and solve.

Example: Solve the system $\begin{cases} y = 2x + 3 \\ 3x + y = 11 \end{cases}$

Step 1: Substitute $y = 2x + 3$ into the second equation.
$3x + (2x + 3) = 11$

Step 2: Combine like terms.
$5x + 3 = 11$
$5x = 8$

Step 3: Solve for x. $x = \frac{8}{5}$

Step 4: Substitute back to find y. $y = 2\left(\frac{8}{5}\right) + 3 = \frac{16}{5} + \frac{15}{5} = \frac{31}{5}$

Answer: $\left(\frac{8}{5}, \frac{31}{5}\right)$

2 SOLVING BY ELIMINATION

Add or subtract the equations to eliminate a variable.

Example: Solve the system $\begin{cases} 2x + 3y = 12 \\ 5x - 3y = 4 \end{cases}$

Step 1: Add the equations to eliminate y.
$\begin{cases} 2x + 3y = 12 \\ +\ 5x - 3y = 4) \end{cases}$
$7x = 16$

Step 2: Solve for x.
$x = \frac{16}{7}$

Step 3: Substitute into either equation to find y.
$2\left(\frac{16}{7}\right) + 3y = 12$
$\frac{32}{7} + 3y = 12$
$3y = \frac{52}{7} \rightarrow y = \frac{52}{21}$

Answer: $\left(\frac{16}{7}, \frac{52}{21}\right)$

3 ELIMINATION (MULTIPLY FIRST)

Sometimes you need to multiply one or both equations first to make the coefficients opposites.

Example: Solve the system $\begin{cases} 3x + 2y = 7 \\ 5x - 4y = 1 \end{cases}$

Step 1: Multiply the first equation by 2 to get $-4y$.
$2(3x + 2y = 7)$
$6x + 4y = 14$

Step 2: Add the equations.
$+\ \ 5x - 4y = 1$
$11x = 15$

Step 3: Solve for x. $x = \frac{15}{11}$

Step 4: Substitute into an equation.
$3\left(\frac{15}{11}\right) + 2y = 7$
$\frac{45}{11} + 2y = 7 \rightarrow 2y = \frac{32}{11} \rightarrow y = \frac{16}{11}$

Answer: $\left(\frac{15}{11}, \frac{16}{11}\right)$

4 TYPES OF SOLUTIONS

A system can have one solution, no solution, or infinitely many solutions.

One Solution (intersecting lines)	**Example:** $\begin{cases} y = x + 1 \\ y = -2x + 4 \end{cases}$	The lines intersect at one point.	**Answer:** (1, 2)
No Solution (parallel lines)	**Example:** $\begin{cases} y = 2x + 3 \\ y = 2x - 1 \end{cases}$	Same slope, different y-intercepts. Parallel lines never intersect.	**Answer:** No solution
Infinitely Many Solutions (same line)	**Example:** $\begin{cases} y = 3x - 4 \\ 2y = 6x - 8 \end{cases}$	Second equation is a multiple of the first. Same line, all points are solutions.	**Answer:** Infinitely many solutions

5 WORD PROBLEM EXAMPLE

Translate the situation into a system of equations, solve, and check if needed.

Example: A school sold 5 adult tickets and 3 student tickets for \$43. Adult tickets cost \$8 more than student tickets. What is the price of each ticket?

Step 1: Define variables. Let s = student ticket price. Then adult ticket price is $s + 8$.

Step 2: Write the system. $\begin{cases} 5(s + 8) + 3s = 43 \\ a = s + 8 \quad \text{(or use } a - s = 8) \end{cases}$

Step 3: Solve.
$5(s + 8) + 3s = 43$
$5s + 40 + 3s = 43$
$8s = 3 \rightarrow s = 3$ Adult price: $3 + 8 = 11$

Answer: Student: \$3, Adult: \$11

6 CHECKING A SOLUTION

Substitute your solution into both equations to make sure it works.

Example: Check $(2, -1)$ for the system $\begin{cases} 3x + y = 5 \\ x - 2y = 4 \end{cases}$

Equation 1:
$3x + y = 5$
$3(2) + (-1) = 5$
$6 - 1 = 5$
$5 = 5$ ✓

Equation 2:
$x - 2y = 4$
$2 - 2(-1) = 4$
$2 + 2 = 4$
$4 = 4$ ✓

Conclusion: $(2, -1)$ is a solution.

7 QUICK REMINDERS & STRATEGY TIPS

DO REMEMBER:
- Choose the method that looks easiest (substitution or elimination).
- Simplify and show your work.
- Check your solution in both equations.
- A system can have one solution, no solution, or infinitely many solutions.
- Use clear variables and equations for word problems.

COMMON MISTAKES:
- Forgetting to multiply before eliminating.
- Making sign errors when adding or subtracting.
- Substituting into the wrong equation.
- Stopping before solving for both variables.
- Not checking the solution.
- Confusing no solution with infinitely many solutions.

TIPS:
- Look for opposite coefficients when using elimination.
- If needed, multiply one or both equations first.
- For word problems, define variables, write a clear system, and solve.
- Always check your answer.